MetroLink Transit Services: Demand Optimization & Service Planning

Data Science Sprint Challenge
CH16 – Public Transport Insight
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Executive analysis of urban bus operations to eliminate bottlenecks and optimize fleet frequency across key city corridors.



Data Preprocessing & Pipeline Integrity

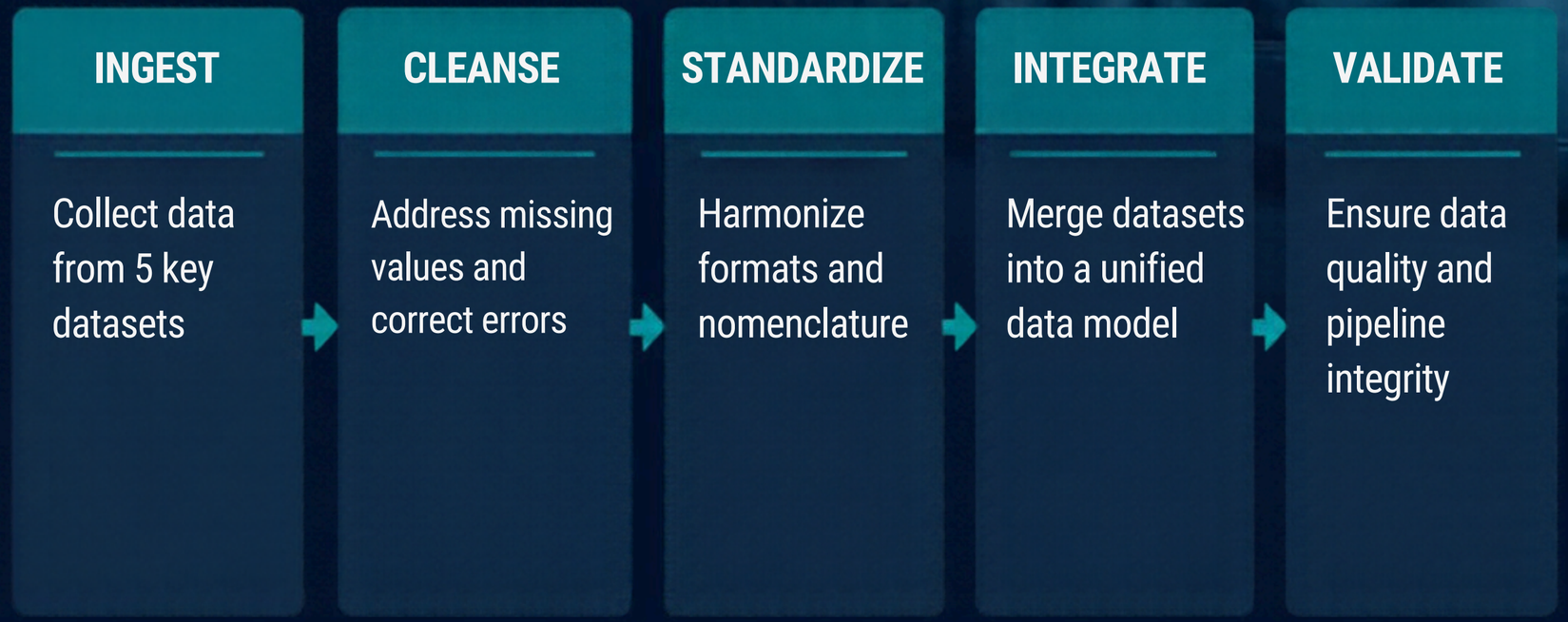
Consolidated **1,200** trip records across 5 datasets: trips, boardings, routes, calendar, operations

Imputed missing stop IDs and corrected negative passenger counts

Removed **40** duplicate boarding logs ensuring clean boarding data

Standardized route nomenclature for consistent analytics

Imputed missing operational delays with median latency values to maintain timeline integrity



Commuter Demand & Temporal Trends

Key insights into ridership patterns that inform service planning

08:00–10:00
17:00–19:00

Bimodal rush hours drive the highest ridership volumes

35%

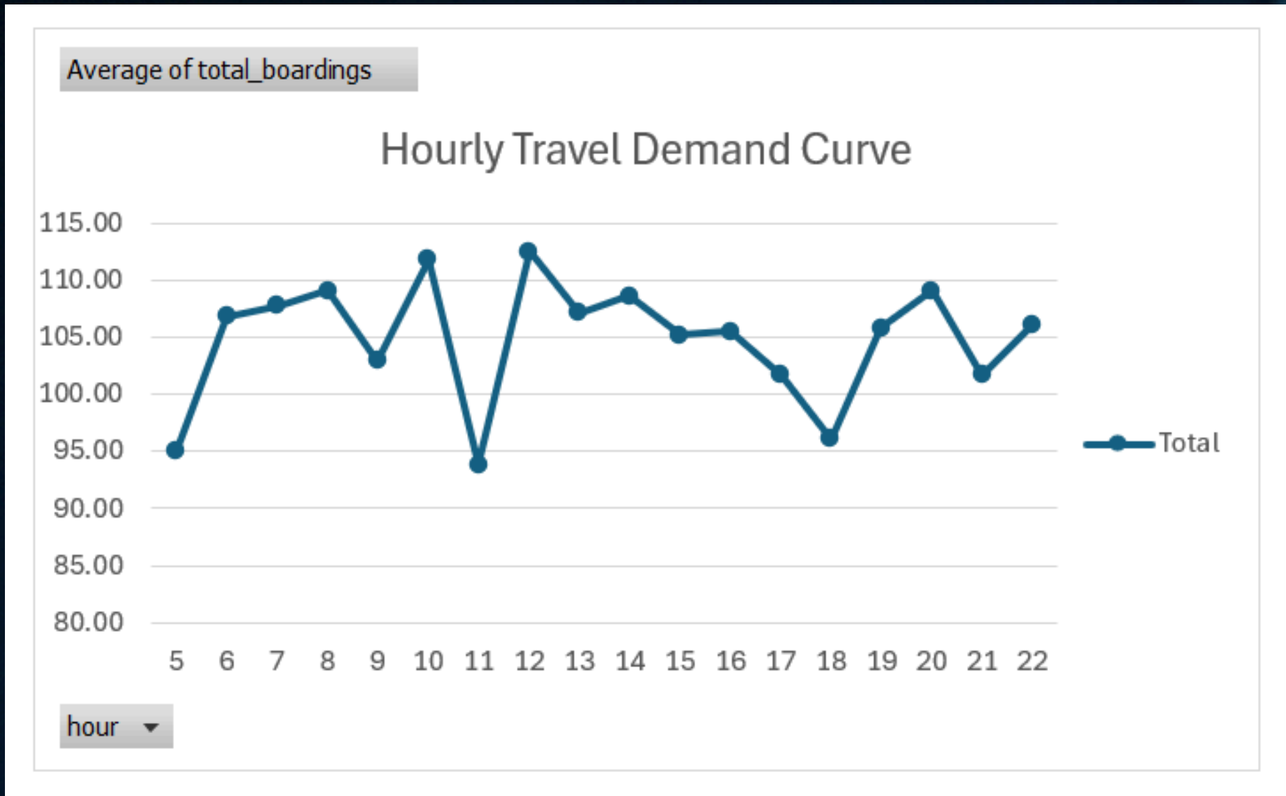
Route R101 (Corridor North) delivers 35% higher volume than suburban routes

20%

Weekday demand is 20% higher than weekends

Special city events spike ridership by 25–35%

RIDERSHIP VOLUME OVER TIME OF DAY



Operational Bottlenecks & Chronic Delays

AVERAGE DELAY

14

MINUTES

on central corridors

ON-TIME RELIABILITY

95%+

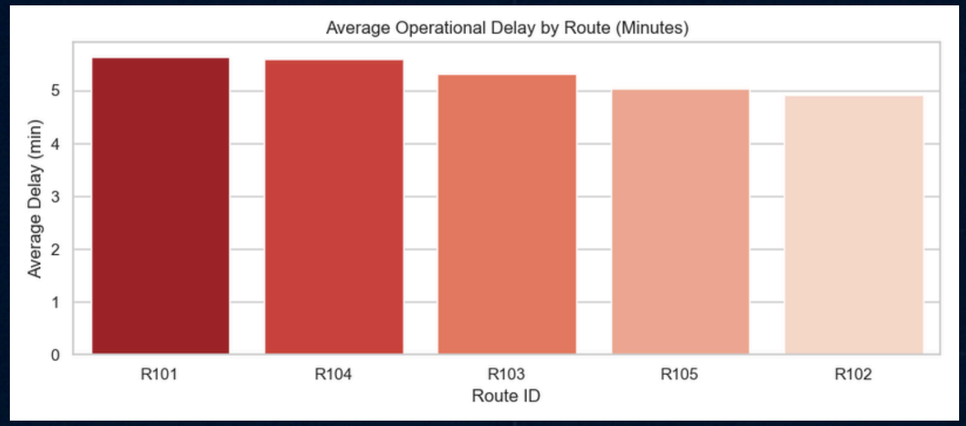
on suburban routes

Central corridors with dense stops face average delays of ~14 minutes due to prolonged boarding dwell times

Suburban routes maintain on-time reliability above 95%

Urgent need identified for curb space management and transit signal priority interventions

DELAY DISTRIBUTION BY CORRIDOR TYPE



Predictive Modeling & Executive Dashboard

MODEL PERFORMANCE

R^2	RMSE
0.86	18.4

Random Forest Regressor model predicting passenger demand

Key predictors: time of day, rush-hour flags, and calendar event schedules

Interactive Excel dashboard deployed with multi-slicer filtering for real-time corridor evaluation

METROLINK DEMAND FORECAST DASHBOARD



Strategic Recommendations & Action Plan

GITHUB URL: MetroLink-Public-Transport-Insight-CH16



DYNAMIC RESCHEDULING

Reallocate idle midday and weekend fleet capacity to peak-hour loops on Route R101 to improve frequency.



EVENT SURGE MANAGEMENT

Pre-dispatch reserve fleets on scheduled event days to proactively manage volume spikes.



INFRASTRUCTURE PRIORITIES

Implement dedicated bus lanes and all-door boarding at top 10 bottleneck stops to reduce delays.